

Data Exploration and Analysis

INA Resources

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CSC 5003

An analysis of the French Media.

Introduction	2
The Dataset	2
The Tools	2
Thematics	3
Theme distribution on different channels: TF1, Arte, Canal+ 2005 - 2020	3
Different themes time evolution through year	4
Parity	10
Women speech time rate per year	10
Women speech time rate per hour	11
Parity between public and private channel	12



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Introduction

For the CSC5003 micro project, we aim to examine the news media coverage and assess the balance between male and females within this sector.

The Dataset

We found interesting data about French media themes and women's speech time on <https://www.data.gouv.fr/fr/datasets/?q=INA>. These datasets gave us interesting information, revealing insights into the themes covered by the media and the proportion of women's speech time across various French channels, broken down by year and by hour.

The CSV files are well-structured and don't need to be cleaned up. There are accessible here:

<https://www.data.gouv.fr/fr/datasets/r/8aba7bac-ddb9-43e1-a257-54d842ad704e>
<https://www.data.gouv.fr/fr/datasets/r/c5b54a7f-2eb2-4db4-a6c5-3b603a56bf29>
<https://www.data.gouv.fr/fr/datasets/r/2d752a77-15d9-46ab-86b4-8a20477593fc>

To compile the dataset, INA utilized a free tool called inaSpeechSegmenter. This tool employs a machine learning algorithm to analyze large volumes of data and detect the voices of both men and women. Regarding thematic classification, the INA algorithm utilizes keywords to categorize the topics covered in TV news.

The Tools

For analysis, we used Scala with Spark for dataframes and applied algorithms to extract relevant information. Visualizations were created with the Breeze library.

The codebase and the graphs generated can be seen on the public Git repository : https://github.com/TheoLardeurGersztein/ina_project.git

Let's explore how French media has changed in the last 20 years in themes and gender balance. This paper will showcase the evolution through simple graphs, providing insights into the results.

Thematics

Theme distribution on different channels: TF1, Arte, Canal+ 2005 - 2020

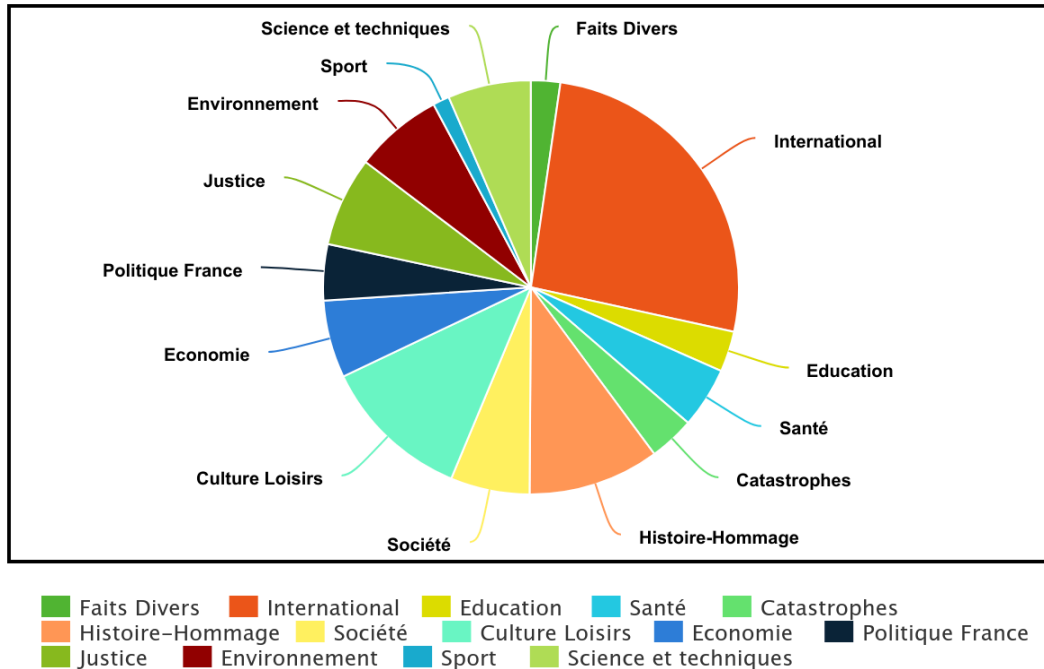


Chart 1: Thematics rate for ARTE

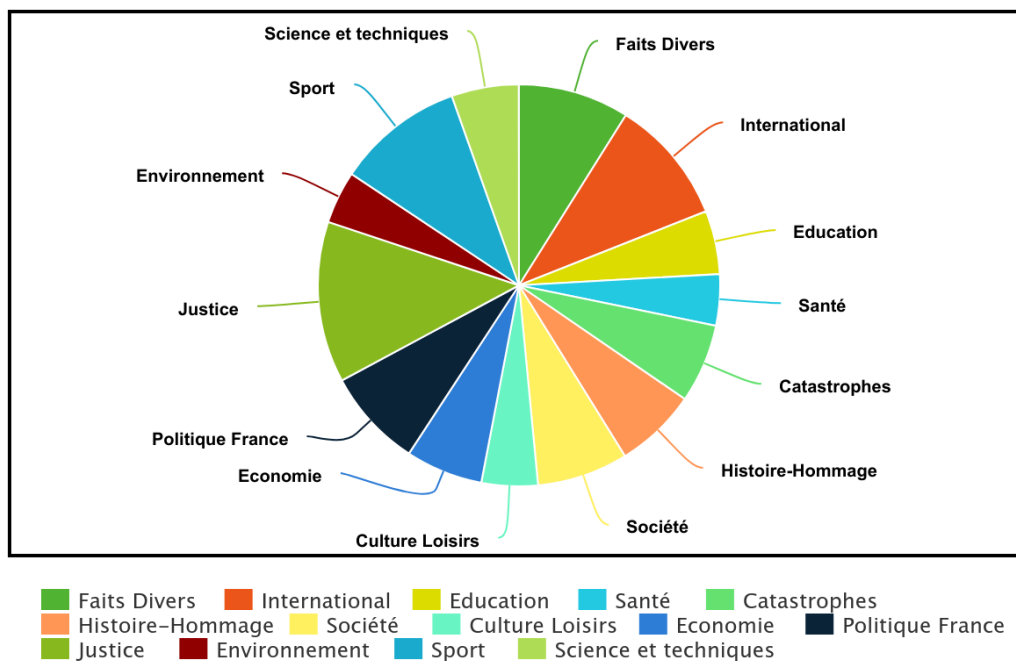


Chart 2: Thematics rate for Canal+

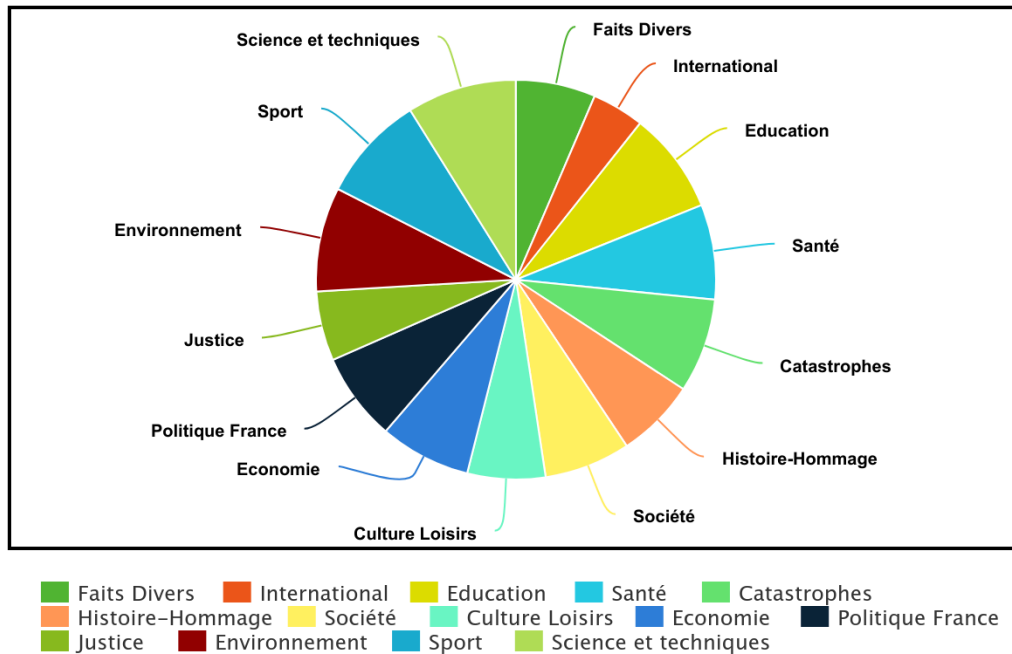
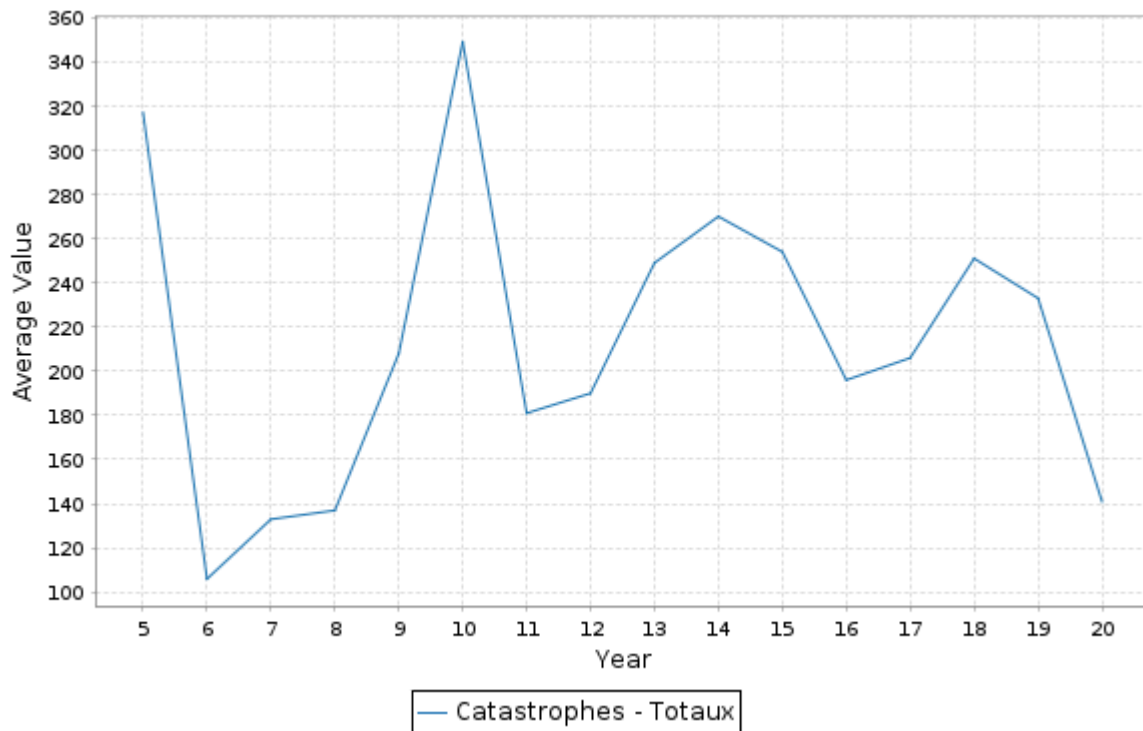


Chart 3: Thematics rate for TF1

Those pie charts show the different topics and thematics that were given in the dataset. Each channel focuses on distinct main topics. Because the data was collected through 15 years, the share of each topic is well balanced but we can see some tendency : TF1 maintains a well-balanced approach, Canal+ leans a bit more towards sports and justice, while ARTE emphasizes international topics, history, and culture. This aligns with the content these media outlets communicate to the public.

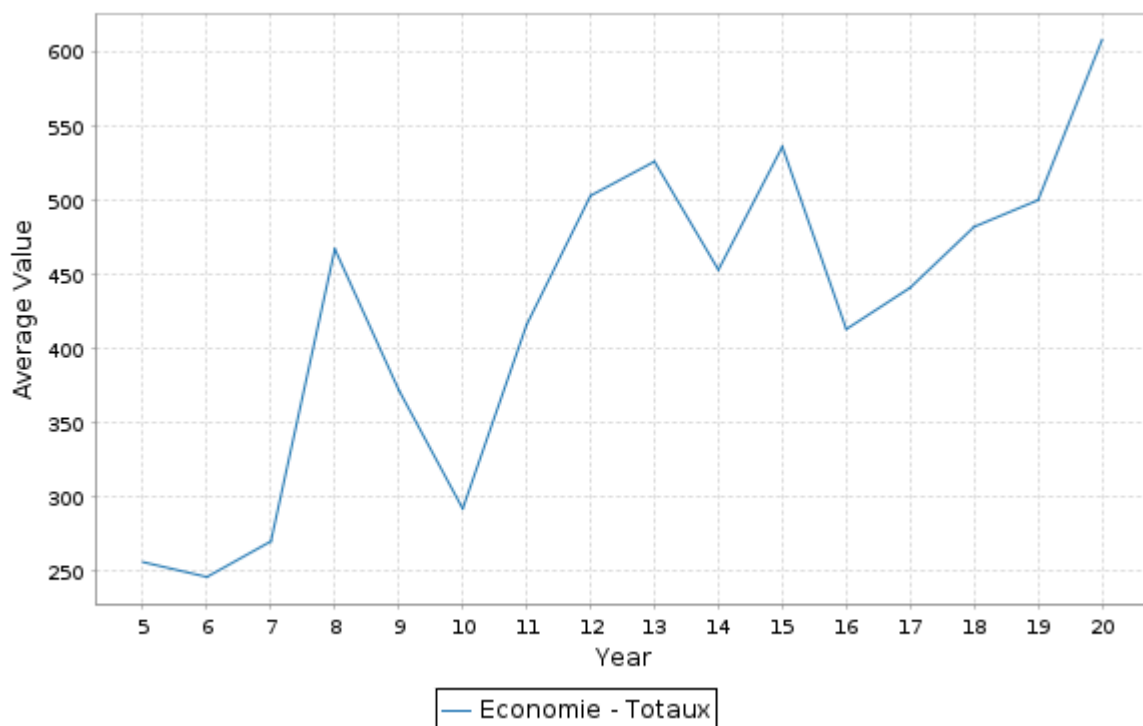
Different themes time evolution through year

Next, we'll delve into various topics frequently covered by the media. Our focus includes Sport, French Politics, Justice, Environment, Economy, Disaster, and Health. Below are the graphs illustrating the evolution of the news coverage on each subject across major French media outlets. On the y-axis, we have the average number of minutes that a specific subject was dealt with over the months and on the x-axis, the years so we can compute the evolution of it.



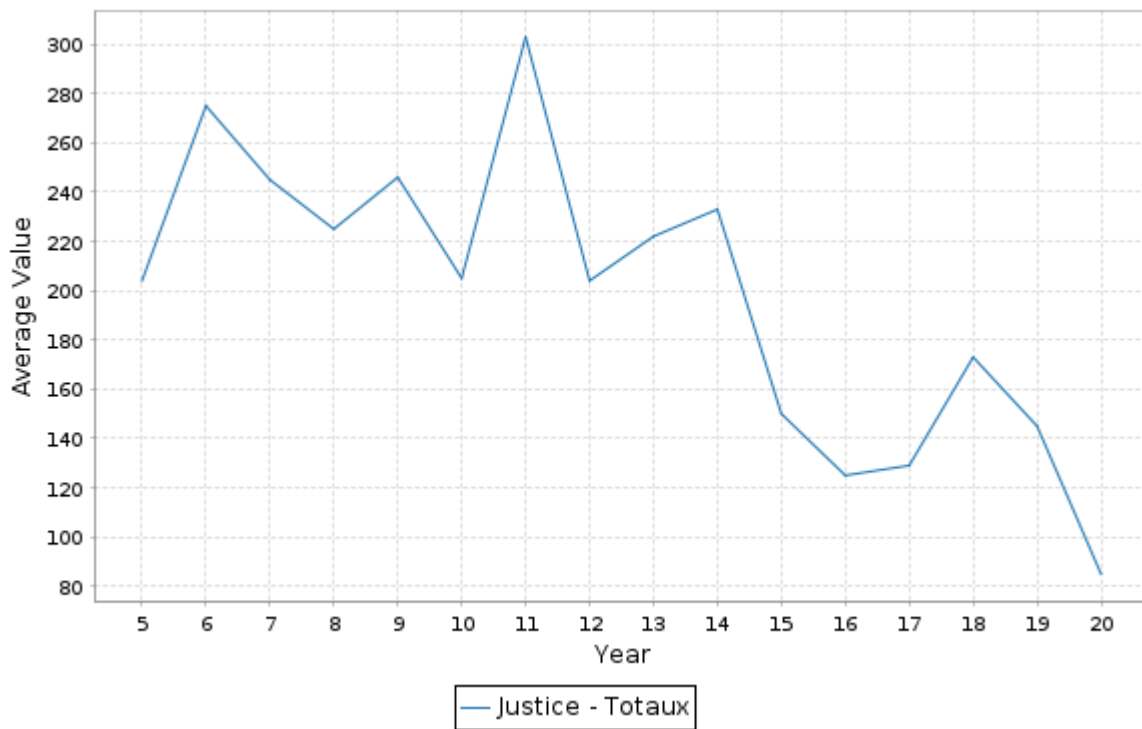
Graph 1: *Disasters*

Graph 1 highlights that the years with the highest media coverage of disasters were 2005 and 2010. This increase can be attributed to significant events such as Hurricane Katrina in 2005 and the earthquake in Haiti in 2010. These major natural disasters likely dominated media discussions during those years, explaining the spike in coverage.



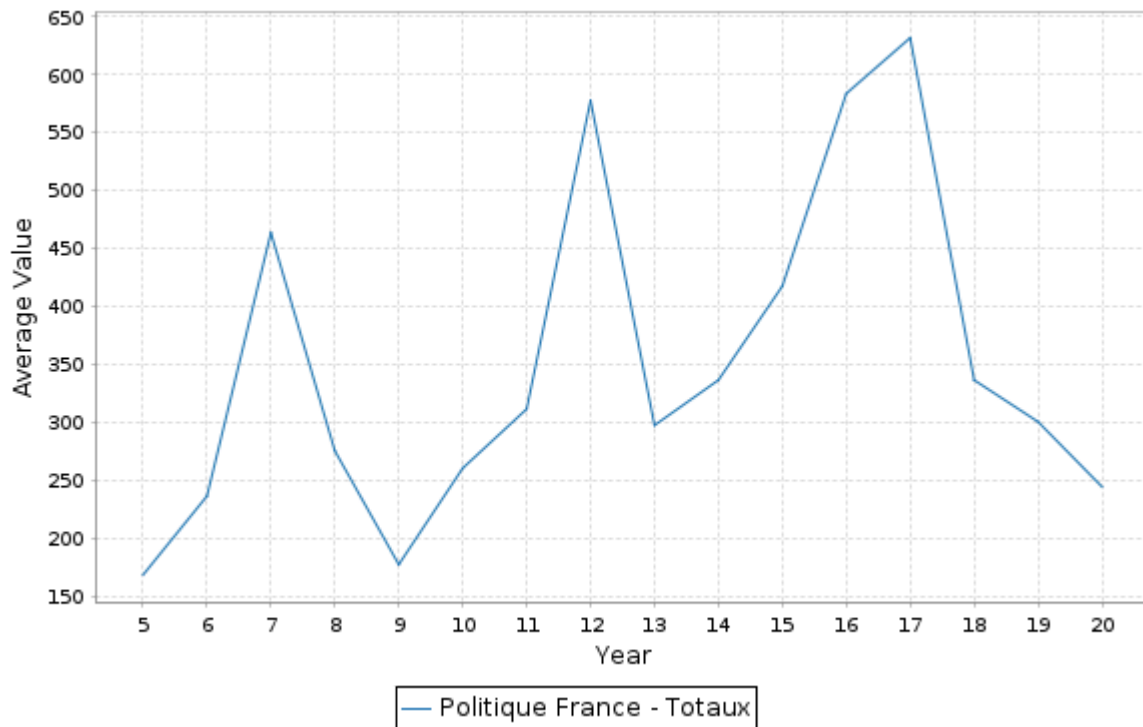
Graph 2: *Economy*

In Graph 2, it's evident that the media coverage of the economy has been on the rise, possibly reflecting growing concerns among the French population about economic matters. The notable increase in 2008 could be attributed to the 2008 subprime economic crisis, which likely heightened media attention on economic issues during that period.



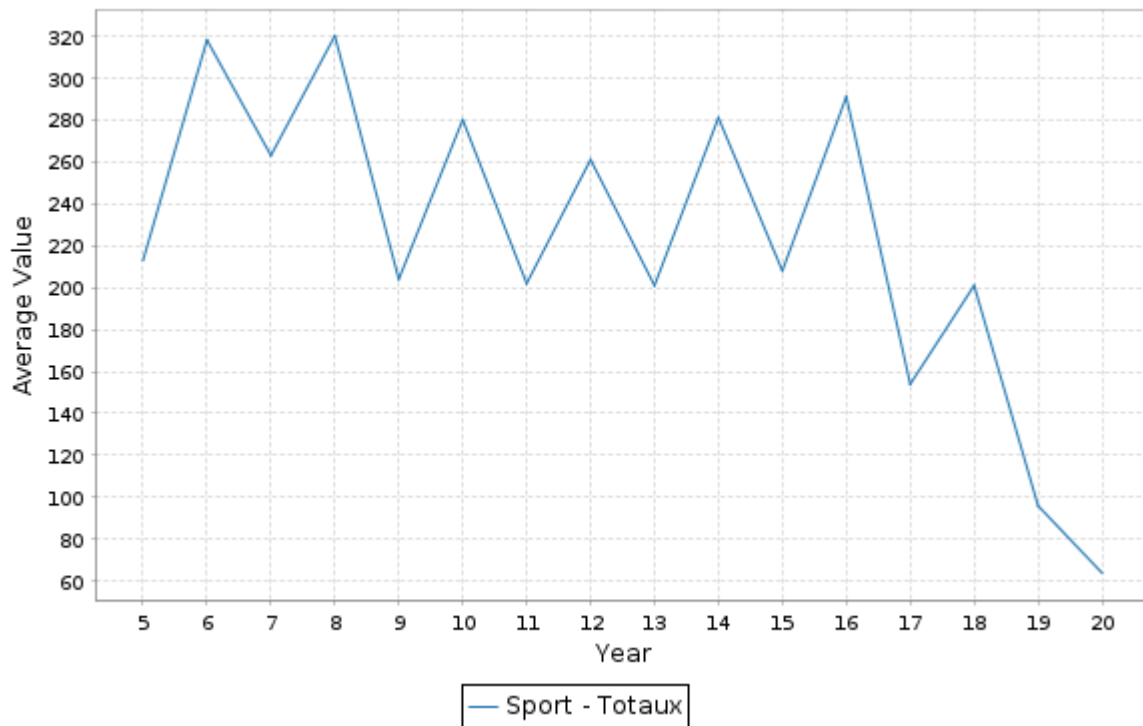
Graph 3: *Justice*

Graph 3 underscores a trend where the topic of Justice appears to be receiving less attention over time. The diminishing coverage suggests a potential shift in media focus away from justice-related issues in recent periods.



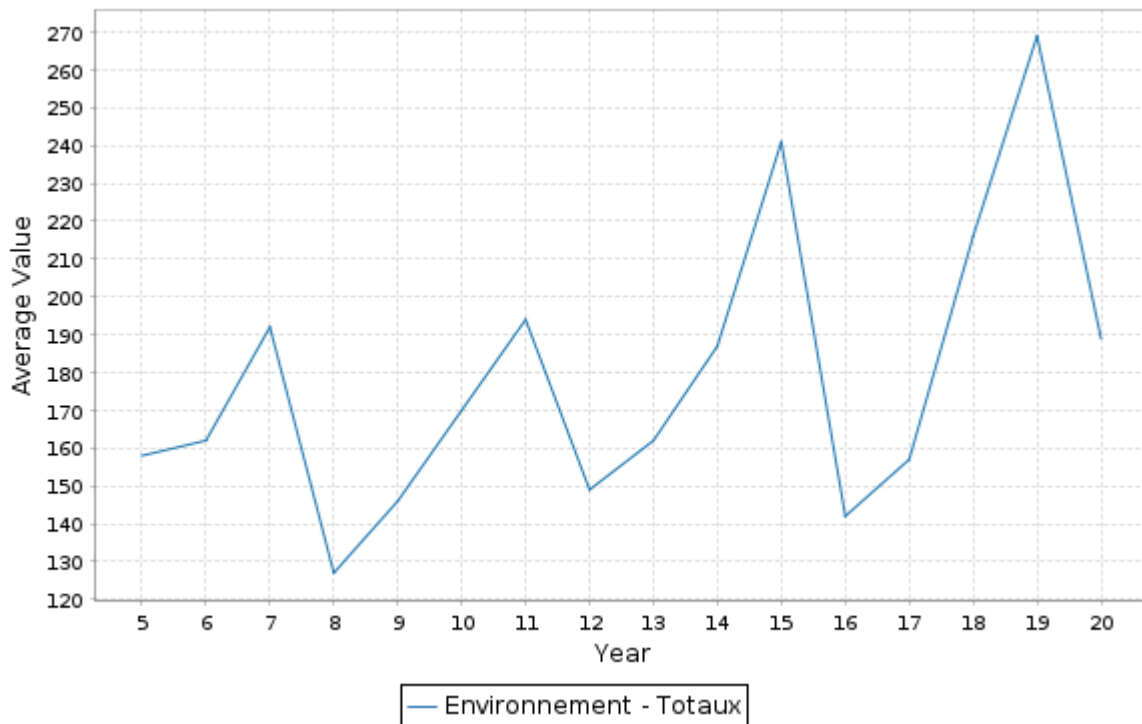
Graph 4: **French politic**

Graph 4 highlights that the media devoted more coverage to French Politics in 2007, 2017, and 2023. These spikes align with the French presidential election years, indicating a notable increase in media attention during these significant political events.



Graph 5: **Sports**

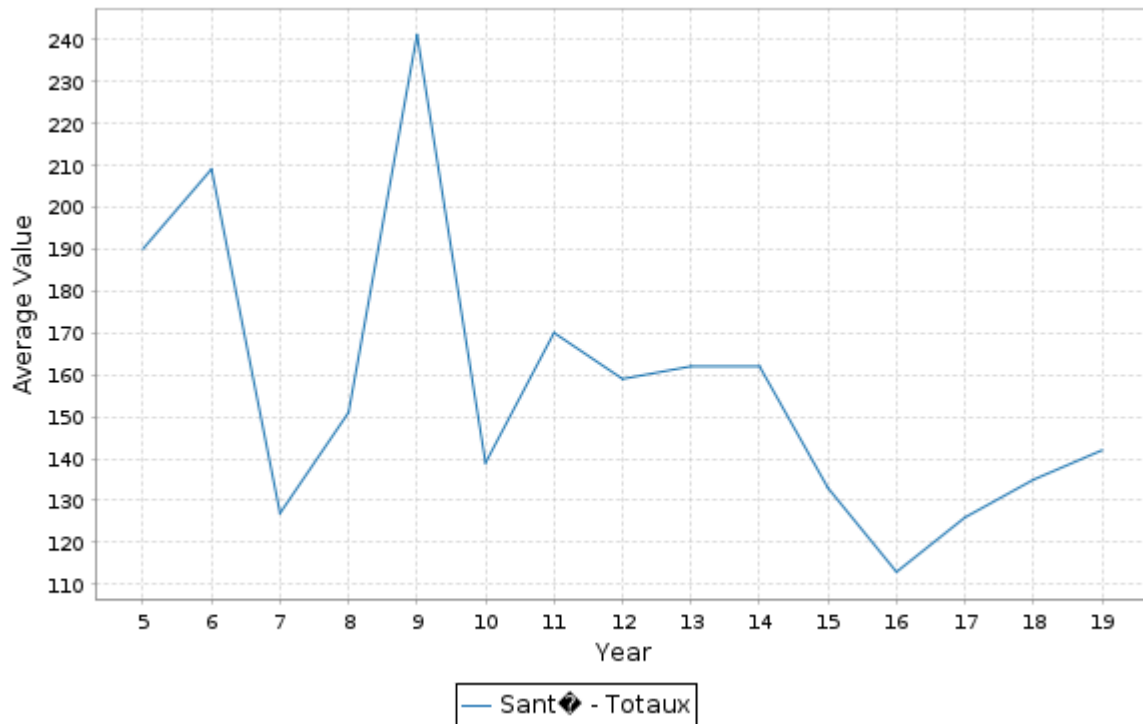
Every two years, there is a noticeable increase in the importance of sports coverage. This pattern corresponds to major international events such as the FIFA World Cup, UEFA Euro, and the Olympic Games, which occur on a biennial cycle.



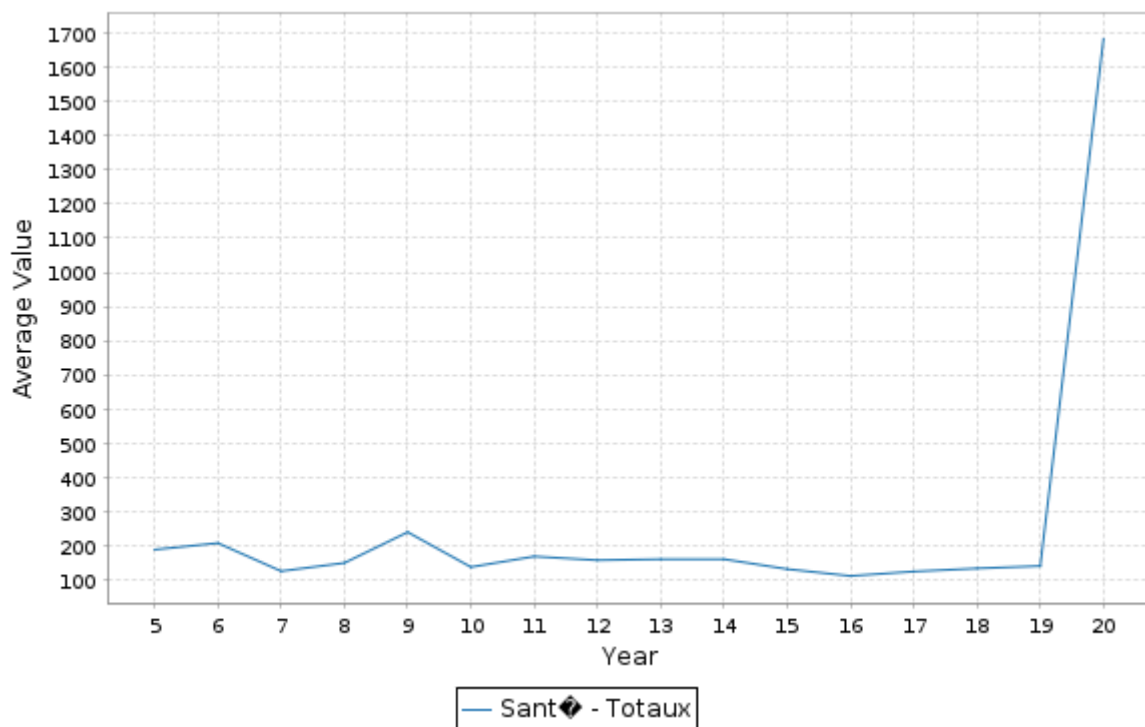
Graph 6: *Environment*

Looking at Graph 6, it becomes apparent that the environment is gaining increasing importance in the media, although the coverage varies significantly. Interestingly, years with no major emphasis on other topics such as sports, politics, or health seem to be the periods when the environment is more prominently discussed in the media.

In the case of Health, there's a distinct deviation due to the significant impact of the COVID-19 pandemic. To provide a clearer perspective, we created two graphs – one excluding 2020 and another including it.



Graph 7: **Health** without 2020



Graph 8: **Health** with 2020

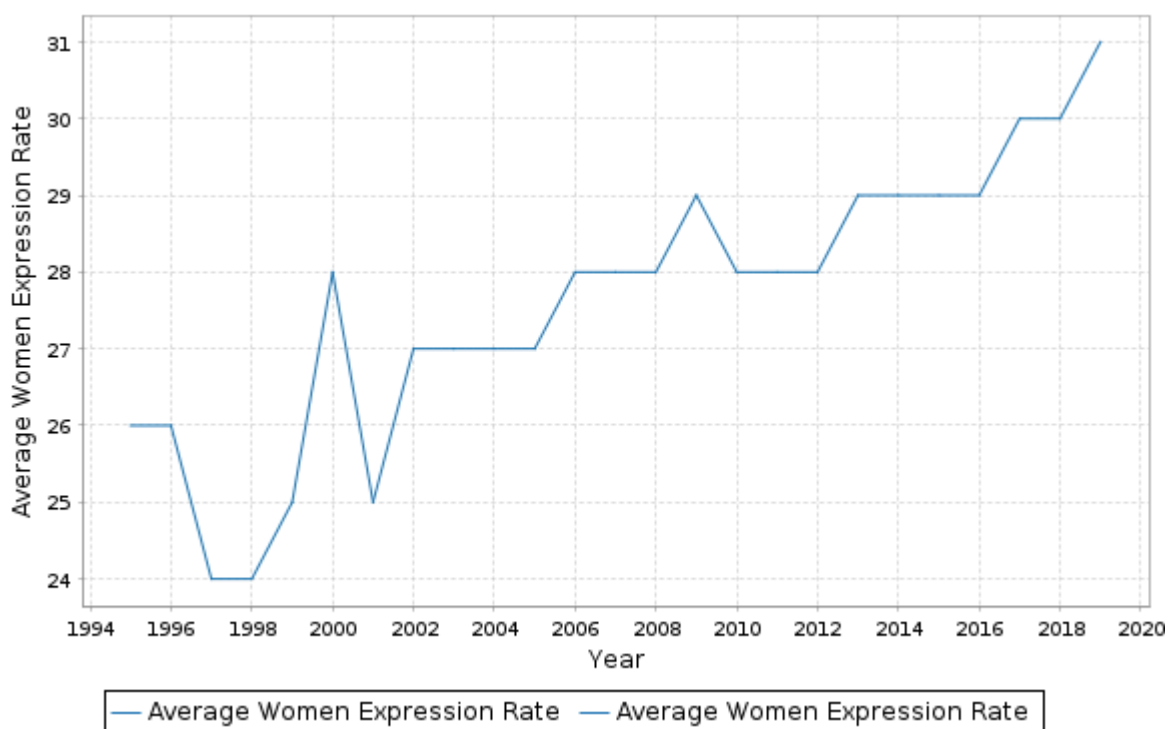
Graph 7 accentuates the H1N1 pandemic in 2009, while Graph 8 reflects the unprecedented surge in health-related coverage attributed to the COVID-19 outbreak in 2020.

A notable observation is that in 2020, there was a decrease in coverage for most subjects such as Sport, Justice, Politics, Environment, and Disaster. In contrast, Health and Economy witnessed an increase, underscoring the predominant concerns of the French media during the context of the COVID-19 pandemic in 2020.

Parity

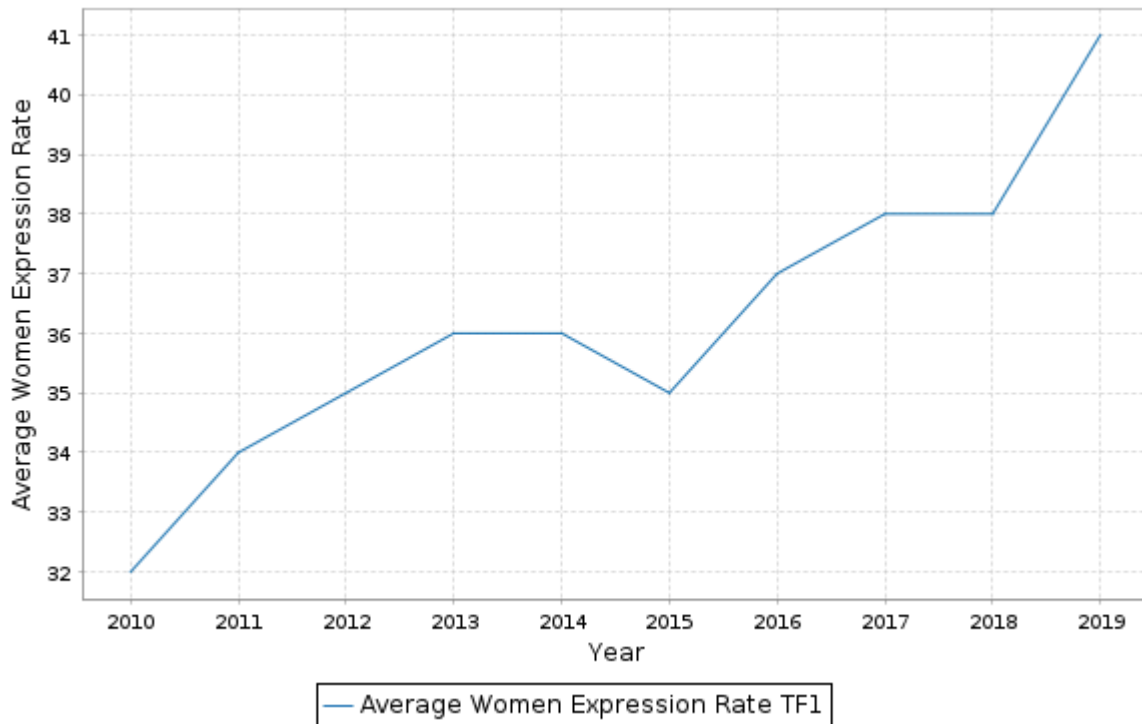
Now, we can delve into the analysis of women's time speech rate across various aspects.

Women speech time rate per year



Graph 9: Average women expression rate each year

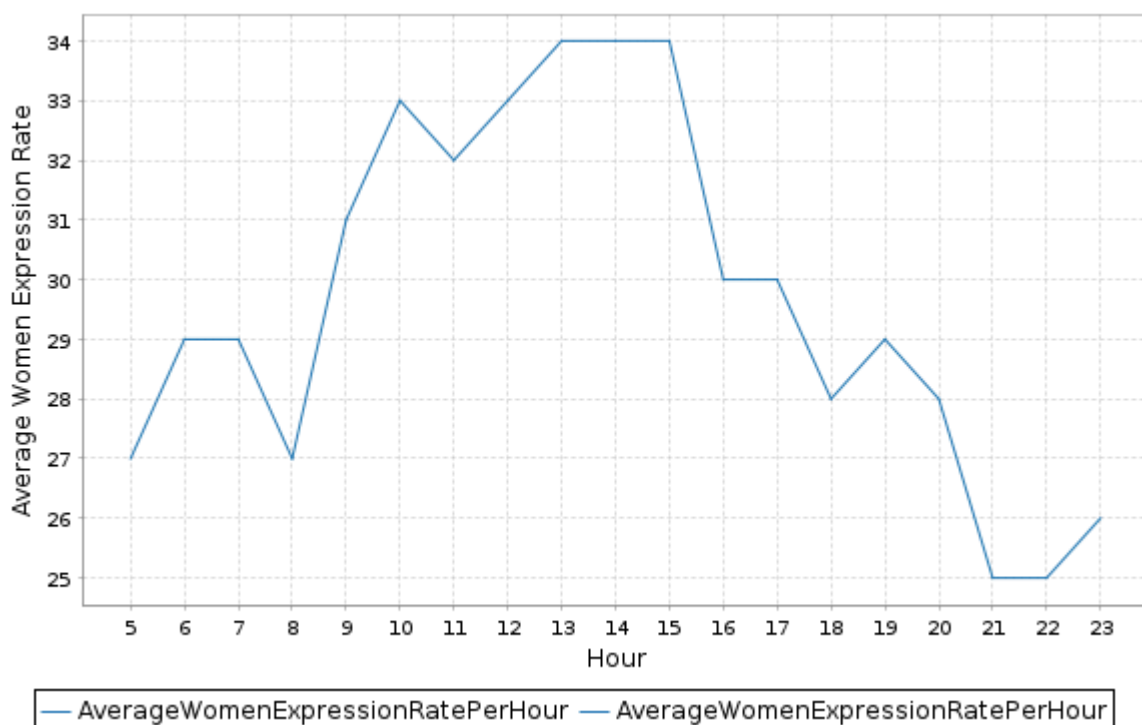
First, let's examine how the rate of women's speech time has evolved over time across all French media. From 1995 to 2020, the rate of women's expression has increased from 25% to 31%. While it remains distant from achieving perfect parity, this upward trend highlights an improvement in women's representation in the media.



Graph 10: Average women expression rate each year in TF1

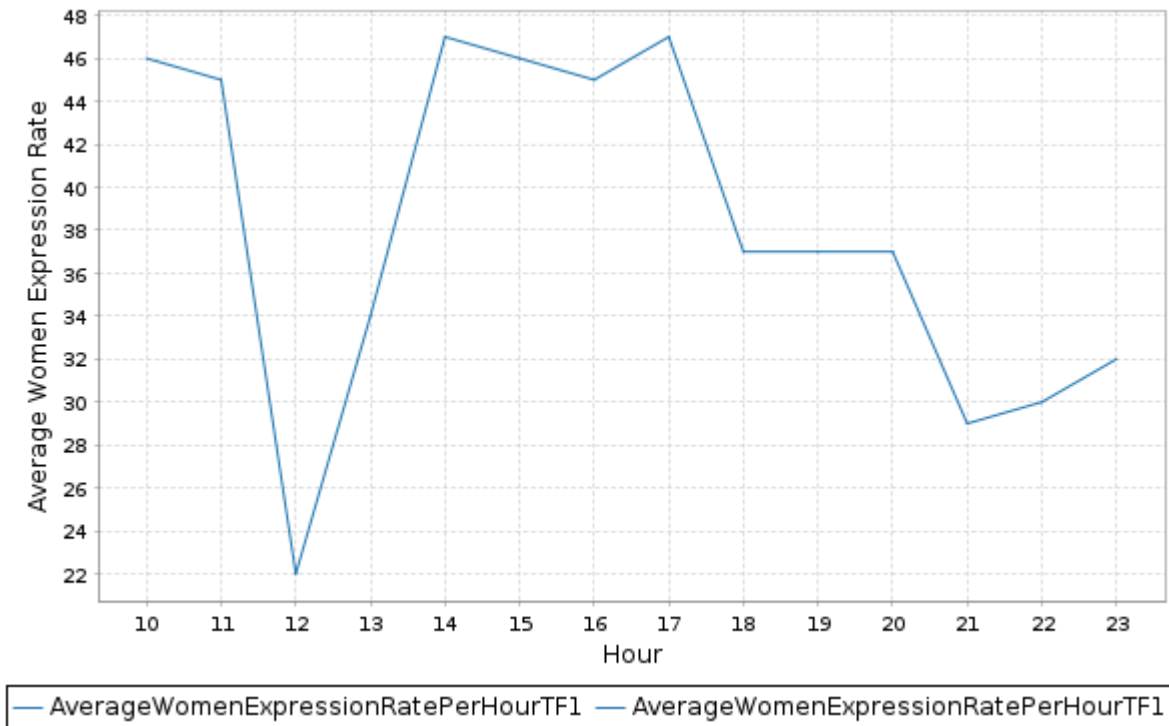
When we narrow our focus to a specific media outlet, such as TF1, we observe on graph 10 that the rate surpasses the overall media average. In 2019, the rate of women's expression on TF1 exceeded 41%, showcasing a notable increase compared to the general trend.

Women speech time rate per hour



Graph 11: Average women expression rate per hour each year

Let's explore the influence of hours on women's speech rate. When considering all media, we observe on graph 11 that the highest rate of women's speech occurs in the morning and from 12 pm to 6 pm, corresponding to off-peak hours. In contrast, the lowest rate is during prime time from 6 pm to 2 am.



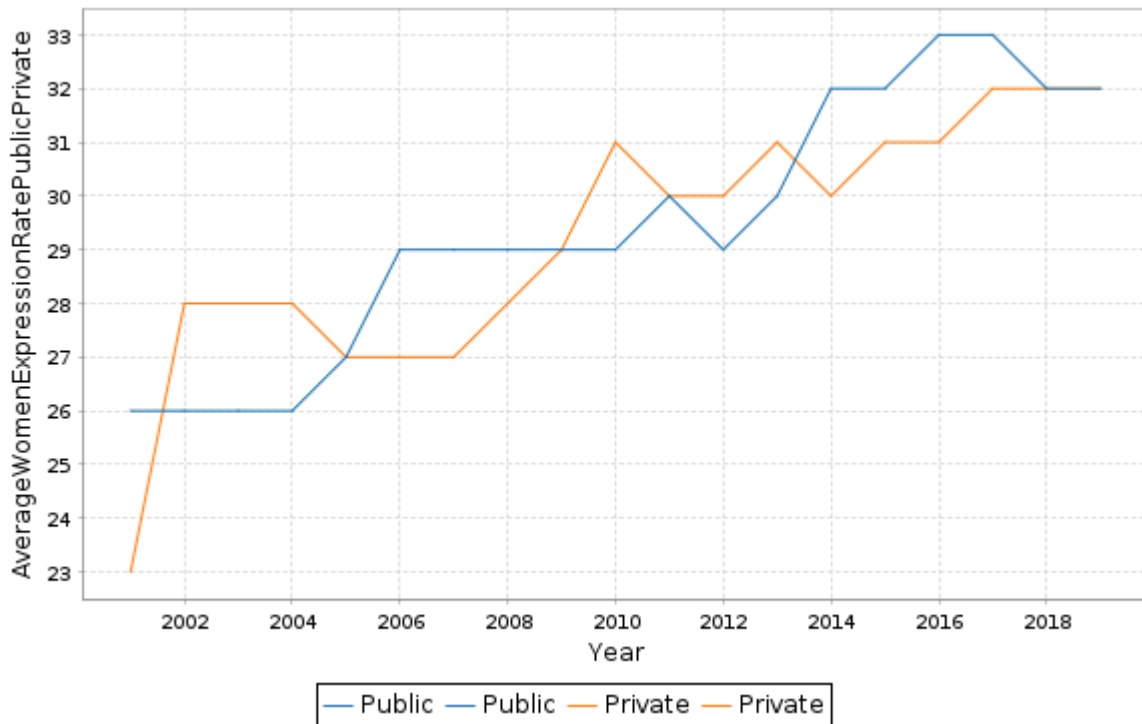
Graph 12: Average women expression rate per hour each year in TF1

Zooming in on TF1, we notice that during off-peak hours, women are almost equally represented, reaching almost 48%. In contrast, during prime time, the representation drops to about 30%. Interestingly, during midday news, around noon, the rate further decreases to 22%.

This evidence suggests that the hours with the highest TV viewership correspond to lower representativity, even for a media outlet like TF1, which generally exhibits good women's representation. The substantial differences observed across hours underscore the impact of timing on gender representation in media.

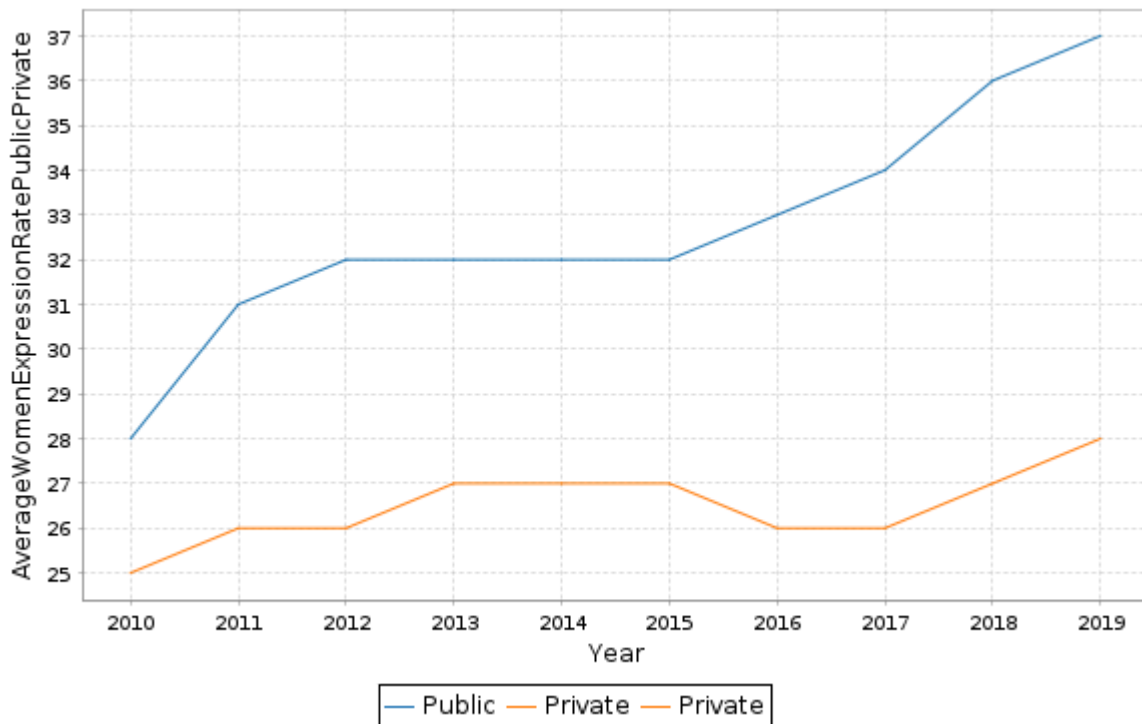
Parity between public and private channel

Now, let's shift our focus to the contrast between public and private channels regarding women's representativity in the media.



Graph 13: Average women expression rate in radio in private and public channels

In the realm of radio, the evolution of women's speech time rate between public and private channels doesn't show a significant difference. While private channels started at a lower level than public ones, they quickly caught up and increased at a similar pace over time.



Graph 14: Average women expression rate in TV in private and public channels

Contrastingly, in the television domain, the dynamics are notably distinct. Private TV channels exhibit marginal increases in women's speech time rate, whereas public channels showcase a substantial and remarkable surge in women's representation over time.

Television demonstrates a more favorable trend in representativity, with a women's speech time rate of 37% in 2019, compared to 32% on radio. This indicates a higher level of women's representation in speech on television as opposed to radio during that year.